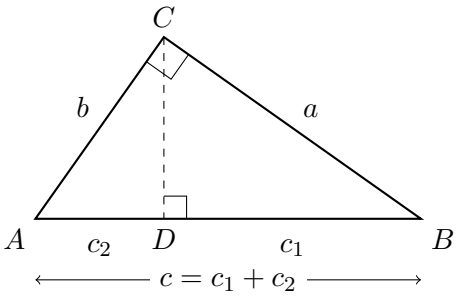
Problem Set 1: Answer Key

Origins of Mathematics, Fall 2026
Worksheet due October 5, 2026

Instructor edition. Accept equivalent correct arguments. Problems 3 and 5 include model responses, not uniquely required wording. Web sources were checked October 13, 2026; numbered citations refer to the source list on the final page.

1. A proof of the Pythagorean theorem

(a) The altitude CD divides the right triangle into two smaller right triangles. The figure uses one representative shape; the proof does not depend on its proportions.



The precise vertex correspondences are

$$\triangle ABC \sim \triangle CBD \sim \triangle ACD.$$

For $\triangle ABC$ and $\triangle CBD$, the angles at C and D, respectively, are right angles, and the angles at B coincide because BD lies along BA. Thus AA similarity applies. For $\triangle ABC$ and $\triangle ACD$, the angles at C and D are right angles, and the angle at A is shared. Hence these triangles are also similar by AA. Consequently the two smaller triangles are similar to each other as well.

(b) From $\triangle ABC \sim \triangle CBD$, corresponding sides give

$$\frac{BC}{AB} = \frac{BD}{CB}, \qquad \text{so} \qquad \boxed{\frac{a}{c} = \frac{c_1}{a}}.$$

From $\triangle ABC \sim \triangle ACD$, corresponding sides give

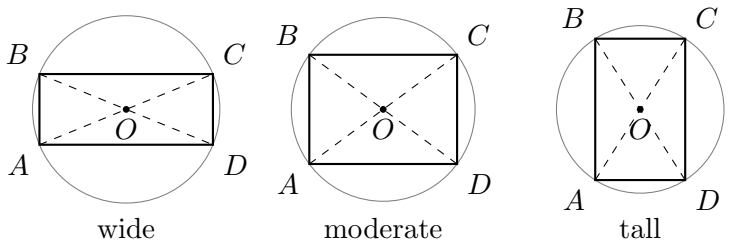
$$\frac{AC}{AB} = \frac{AD}{AC}, \qquad \text{so} \qquad \boxed{\frac{b}{c} = \frac{c_2}{b}}.$$

(c) All lengths are positive. Clearing denominators yields $a^2 = cc_1$ and $b^2 = cc_2$. Therefore

$$\boxed{a^2 + b^2 = cc_1 + cc_2 = c(c_1 + c_2) = c^2}.$$

2. An angle in a semicircle

(a) Three rectangles with different proportions are shown below. In each, O is the intersection of the diagonals.

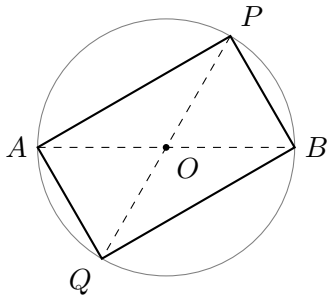


A rectangle is a parallelogram, so its diagonals bisect each other. Its diagonals are also equal (for example, by SAS congruence of right triangles formed from two adjacent sides). Thus

$$OA = OC = \tfrac{1}{2}AC = \tfrac{1}{2}BD = OB = OD.$$

All four vertices lie on the circle centered at O with this common radius. Both AC and BD are diameters: they pass through O and have endpoints on the circle.

(b) Let O be the center of the given circle. For an arbitrary point P on the semicircle, other than A or B, construct its antipodal point Q, so that PQ is a diameter. Join the vertices in the order A, P, B, Q.



The diagonals of $APBQ$ are AB and PQ. They bisect each other at O, so $APBQ$ is a parallelogram. They are equal, since both are diameters of the same circle. A parallelogram with equal diagonals is a rectangle. To justify the latter fact without Thales' theorem, compare triangles APB and PAQ: AP is shared, $PB = AQ$ (opposite sides), and $AB = PQ$ (diagonals). SSS gives equality of the adjacent interior angles at P and A. Adjacent angles of a parallelogram sum to $180°$, so both equal $90°$. Hence $\boxed{\angle APB = 90°}$.

(c) *Sample reflection.* The rectangles suggest completing the inscribed triangle by adding the point opposite P. The picture helps identify the useful diagonals, but their bisection and equality must be justified, as must the rectangle criterion. The argument works for every allowed P because the antipodal point always exists and the two diameters always have the same midpoint and length. Three drawings alone would not establish this general statement.

3. What can we learn about Thales from later authors?

Model essay (approximately 470 words). Other carefully documented passages and evaluations are acceptable.

Aristotle and Proclus offer evidence about the later reputation of Thales, but neither was an eyewitness. Thales lived approximately 624–547 BCE. Aristotle wrote in the fourth century BCE, roughly two centuries after Thales' death. Proclus lived in 411–485 CE, about a millennium after Thales. These intervals do not automatically discredit their reports, but they make the transmission of information an important question. The approximate chronology is supported by the editions and biographical references below. [1–4]

In *Metaphysics* I.3, 983b20–27, translated by Hugh Tredennick, Aristotle identifies water as Thales' fundamental principle and connects this doctrine with the claim that the earth rests on water. He suggests that observations about moisture, nourishment, heat, and seeds may explain the doctrine. Tredennick's word "Presumably" matters: the explanation of Thales' reasoning is conjectural, even though the attribution of the water doctrine is presented more directly. Aristotle names no source for this particular report. His surrounding discussion organizes earlier thinkers according to the kinds of causes they recognized. Consequently, his account is evidence for how he understood Thales within his own philosophical investigation, rather than a transcript of Thales' words. The doctrinal attribution has stronger textual support here than the reconstructed motivation, but the passage alone cannot independently verify either historical claim. [1]

In his commentary on Euclid's definition of a diameter, Proclus reports that Thales first demonstrated that a diameter bisects a circle. I use Thomas Taylor's translation, volume I, commentary on Definition XVII, pages 165–166. The wording "they report" presents the attribution as received information; no earlier authority is named in this particular sentence. Proclus then supplies a mathematical explanation involving the two halves of the circle. We must distinguish that explanation from the historical assertion about Thales. Its position after the attribution does not establish that it reproduces Thales' own proof. Proclus is explaining geometry, so his passage can simultaneously illuminate a mathematical claim and preserve a tradition about its discoverer. Its late date and unnamed immediate source limit our ability to establish the priority claim. It nevertheless directly documents that this attribution circulated by Proclus' time. [2]

A successful reconstruction and a historical identification therefore answer different questions. A reconstruction asks whether stated geometric facts can establish a theorem. Historical identification asks whether a particular person actually used that chain of reasoning. Problem 2 gives a valid rectangle construction for any permitted point on a semicircle. It shows how the theorem can be proved from suitable geometric properties; it does not by itself show that those properties were available to Thales, still less that he used this construction. That stronger conclusion would require additional historical evidence connecting Thales with the method. Neither selected passage supplies that connection.

What to look for when grading: identifiable passages and editions; sensible approximate time gaps; attention to named or unnamed sources; a distinction between explicit statements and inferences; and a clear separation of mathematical validity from historical attribution. Students need not choose these same passages or reach identical assessments of their reliability.

4. Parity in Pythagorean triples

(a) Let m be an arbitrary integer. Every integer is even or odd.

- If $m = 2k$ for an integer k, then $m^2 = 4k^2$, so its remainder on division by 4 is 0.
- If $m = 2k + 1$ for an integer k, then

$$m^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1,$$

so its remainder on division by 4 is 1.

Thus **every integer square has remainder** 0 **or** 1 **modulo** 4.

(b) Suppose for contradiction that both a and b are odd. By part (a), write $a^2 = 4r+1$ and $b^2 = 4s+1$ for integers r, s. Then

$$c^2 = a^2 + b^2 = 4(r + s) + 2.$$

This makes c^2 have remainder 2 when divided by 4, contradicting part (a). Therefore a **and** b **cannot both be odd**.

Optional challenge. Let (a, b, c) be primitive. Part (b) shows that at least one leg is even. If both legs were even, then $a^2 + b^2 = c^2$ would be even. An odd integer has an odd square, so c would have to be even too. Then 2 would divide all three entries, contradicting primitivity. Hence exactly one of a, b is even and the other is odd.

The square of the even leg is even, and the square of the odd leg is odd. Their sum c^2 is odd. Since the square of an even integer is even, c must be odd. Therefore

Exactly one leg is even, the other leg is odd, and the hypotenuse is odd.

Common gap: It is not enough to say that a primitive triple "cannot contain even numbers." It can contain an even leg; what primitivity rules out here is a common factor of 2 in all three entries.

5. Mathematics in progress

(a) *Model response (approximately 240 words).*

GIMPS searches for primes of the form $2^p - 1$ by distributing computational jobs among volunteers' computers. Its PrimeNet system coordinates assignments and records results, allowing many machines to contribute to a shared search. Participants run software such as Prime95 or suitable GPU software. Their contributions include eliminating candidates, testing remaining numbers, and checking earlier computations. Account information can track computers, assignments, and completed work. [5,6]

The search concentrates on prime exponents p: if p is composite, $2^p - 1$ cannot be prime. A prime exponent, however, does not guarantee a prime result. The project first tries economical factoring methods, including trial factoring and a method called $P - 1$. Finding a nontrivial factor proves that a candidate is composite and avoids a much longer test. Failure to find a factor does not prove primality. Candidates that survive receive a probable-prime test. A composite result rules out that candidate; a probable-prime result calls for a definitive Lucas–Lehmer test. [6,7]

Reliability is essential because long computations can suffer errors. Modern probable-prime runs can produce proof files that other computations verify with much less work than repeating the entire test. These verify the computation; a probable-prime outcome still requires primality confirmation. Older results may need independent double-checks, and a potential record receives independent confirmation on different hardware and programs. Thus a participant who never discovers a prime still contributes useful evidence: their machine reduces the search space or increases confidence in earlier results. The project combines many separate contributions rather than requiring one computer to conduct the entire search. [7–9]

(b) As reported by GIMPS when checked on **October 13, 2026**, the largest known prime is

$$\boxed{2^{136{,}279{,}841} - 1}.$$

- **Discovery date: October 12, 2024.** GIMPS credits Luke Durant with finding it on this date. [9]
- A probable-prime result appeared on October 11; a Lucas–Lehmer test confirmed primality on October 12. Further independent confirmations concluded on October 19. [8,9]
- The public announcement was October 21, 2024. This is not the discovery date. [9]
- The number has 41,024,320 decimal digits. [9]

Grading note. This is a date-sensitive answer. Evaluate a student's response against the source and check date they report. A dated website announcement should not be confused with the date of discovery. Optional participation is not required for credit; no participation is needed to answer (a) or (b).

6. Triangular numbers and squares

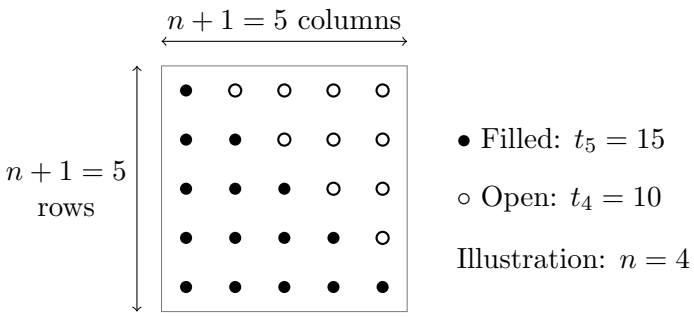
(a) Add only the new last term:

$$\boxed{t_{1001} = t_{1000} + 1001 = 500500 + 1001 = 501501}.$$

(b) We have $t_1 + t_2 = 1 + 3 = 4 = 2^2$, $t_2 + t_3 = 3 + 6 = 9 = 3^2$, and $t_3 + t_4 = 6 + 10 = 16 = 4^2$. The conjecture is

$$\boxed{t_n + t_{n+1} = (n+1)^2 \qquad (n \geq 1)}.$$

(c) Use an $(n+1)$-by-$(n+1)$ square array. In row r, counting from the top, place r filled dots on the left and $n + 1 - r$ open dots on the right.



The filled dots have row lengths $1, 2, \ldots, n+1$, so they form a triangular array with t_{n+1} dots. The open dots have row lengths $n, n-1, \ldots, 1, 0$, forming a rotated triangular array with t_n dots. Every row contains exactly $n+1$ dots; no position is omitted or counted twice. Thus the two arrays form a square with $(n+1)^2$ dots. This construction works for every positive integer n.

(d) Algebraically,

$$\begin{aligned} t_n + t_{n+1} &= \frac{n(n+1)}{2} + \frac{(n+1)(n+2)}{2} \\ &= \frac{(n+1)(n+n+2)}{2} = (n+1)^2. \end{aligned}$$

The two summands count the two triangular parts, and the final square counts the entire array.

(e) Three cases leave infinitely many positive integers unchecked. The proof in (c) gives a rule for an arbitrary n and accounts for every position of the square. Its validity does not depend on the illustrative choice $n = 4$.

(f) Remove the terms 1 through 138 from the sum 1 through 296:

$$\begin{aligned} 139 + \cdots + 296 &= t_{296} - t_{138} \\ &= \frac{296 \cdot 297}{2} - \frac{138 \cdot 139}{2} \\ &= 43{,}956 - 9{,}591 = \boxed{34{,}365}. \end{aligned}$$

There are $296 - 139 + 1 = 158$ terms; pairing first and last gives $79 \cdot 435 = 34{,}365$ as a check.

Sources and passage locators

All web sources checked October 13, 2026. Links open the consulted texts or project pages.

[1] Aristotle, *Metaphysics*, I.3, **983b20–27**. Translated by Hugh Tredennick, *Aristotle in 23 Volumes*, vols. 17–18 (Harvard University Press / William Heinemann; Perseus edition identifies 1933, 1989). Read the passage in Perseus. For fourth-century dating, see also the Internet Classics Archive edition information (W. D. Ross translation; approximate date 350 BCE).

[2] Proclus, *The Philosophical and Mathematical Commentaries on the First Book of Euclid's Elements*, translated by Thomas Taylor, vol. I (London, 1792), **commentary on Definition XVII, pp. 165–166**. Project Gutenberg text. Locate "DEFINITION XVII" and the sentence beginning "But they report, that Thales". Definition numbering follows this edition; it is the definition of a circle's diameter.

[3] J. J. O'Connor and E. F. Robertson, *Thales of Miletus*, MacTutor History of Mathematics, University of St Andrews. Biography and approximate dates. Secondary reference for chronology and the limitations of the surviving evidence.

[4] J. J. O'Connor and E. F. Robertson, *Proclus*, MacTutor History of Mathematics, University of St Andrews. Biography and dates. Secondary reference for the fifth-century setting.

[5] GIMPS, Getting Started on GIMPS, especially account information, software, assignments, and results.

[6] GIMPS, How GIMPS Works, "Overview" and work-type table; page updated April 3, 2026.

[7] GIMPS, Mathematics and Research Strategy, especially "Overview", "PRP (Probable Prime) Testing", "Lucas-Lehmer Primality Testing", and "Double-Checking"; page updated October 16, 2024.

[8] GIMPS, home page, current record announcement and the news items "First-time Lucas-Lehmer Testing Ends" (April 8, 2021) and "BIG Changes Are Here! Prime95 version 30.3 released" (September 10, 2020), concerning PRP proofs and verification.

[9] GIMPS, 52nd Known Mersenne Prime Discovered, press release dated October 21, 2024; see the opening announcement and "Verifying the new prime" for the discovery and confirmation dates.